Computer-Assisted Short Answer Grading Using Large Language Models and Rubrics

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Abstract: Grading student answers and providing feedback are essential yet time-consuming tasks for educators. Recent advancements in Large Language Models (LLMs), including ChatGPT, Llama, and Mistral, have paved the way for automated support in this domain.

This paper investigates the efficacy of instruction-following LLMs in adhering to predefined rubrics for evaluating student answers and delivering meaningful feedback. Leveraging the Mohler dataset and a custom German dataset, we evaluate various models, from commercial ones like ChatGPT to smaller open-source options like Llama, Mistral, and Command R. Additionally, we explore the impact of temperature parameters and techniques such as few-shot prompting.

Surprisingly, while few-shot prompting enhances grading accuracy closer to ground truth, it introduces model inconsistency. Furthermore, some models exhibit non-deterministic behavior even at near-zero temperature settings. Our findings highlight the importance of rubrics in enhancing the interpretability of model outputs and fostering consistency in grading practices.

Keywords: Natural Language Processing, Automatic Short Answer Grading, Large Language Models, Rubrics, Mistral, ChatGPT, Llama

1 Introduction

Grading student answers is crucial in education, requiring fairness, efficiency, consistency, and fostering student learning through meaningful feedback. However, the manual assessment process is often subjective, labor-intensive and creates bottlenecks in educational workflows.

Automatic Short Answer Grading (ASAG) aims to address these challenges by assisting educators in grading short answers, typically one-paragraph long. With the recent advancements in instruction-following large language models (LLMs), these models are increasingly considered for grading. Although they often demonstrate promising performance, they also face challenges such as inconsistency, bias, and hallucinations. Rubrics could enhance the interpretability of these models' outputs.

Rubrics serve as standardized frameworks for consistently grading student answers. By providing clear (yes/no) criteria for assessment, rubrics help reduce inconsistency and bias

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doi:10.18420/inf2024_121

in grading, while also serving as feedback mechanisms, ultimately enhancing students' understanding and acceptance of their grades.

In this study, we explore the extent to which instruction-following models can adhere to and benefit from rubrics when evaluating student answers. Additionally, we examine the consistency of model outputs and investigate the potential benefits of few-shot prompting, wherein graded answers are provided as examples to the models. In our study, we compare commercial online models with models that can be run locally, addressing potential privacy concerns. We deliberately opt for smaller models to minimize additional hardware requirements.

2 Background & Related Work

Transformer-based language models [De19] utilizing the attention mechanism [Va17] have significantly advanced the field of natural language processing. OpenAI demonstrated that LLMs can achieve state-of-the-art performance across various domains when provided with examples [Br20]. Additionally, Meta introduced LLMs with open licenses which allow practitioners to run models locally [To23].

Automated Short Answer Grading (ASAG) has seen growing interest recently due to advancements in natural language processing and machine learning [BGS15; BRK21]. Schneider et al. [Sc23] evaluated large language models (LLMs) for autograding, noting that while these models offer valuable support, they still require human oversight for independent grading. Chang and Ginter [CG24] investigated ASAG for Finnish courses using ChatGPT. They found GPT-4 outperformed GPT-3.5, and one-shot outperformed zero-shot approaches, with fewer errors in the output. Yoon et al. [Yo23] compared one-shot prompting and text similarity models using rubrics, finding fine-tuned BERT performed best. Jiang [JB24] showed that few-shot prompting improves accuracy and QWK, with combined score and rationale approaches being most effective.

Apart from these studies on the applicability of LLMs to ASAG, a meta-study by Panadero et al. [Pa23] showed that rubrics, such as those discussed by Walvoord [WSL13] and Allen [AT06], have a positive effect on academic performance.

3 Experiments

In this section we describe our process for evaluating if instruction following models are able to grade short answers, and if the performance of the model increases when provided with predefined grading rubrics. We investigate the consistency of model outputs and the influence of few shot prompting on the quality of the results.

3.1 Choice of Models

We chose 5 currently common and high-ranking Large Language Models for our experiments (see Table 1): Two proprietary models from OpenAI and three models that can be run locally. For the selection of the local models, their size was taken into consideration in order to be able to run them on the existing hardware (as described in the next section). Additionally, GPT 4o was only used for the second, smaller dataset due to the trade-off between using a common model and the associated cost.

Model	Organization	License	required GPU Ram
GPT 3.5	OpenAI	Proprietary	-
GPT 4o	OpenAI	Proprietary	-
Mistral 7B Instruct v0.2	Mistral AI	Apache 2.0	30 GB
Command R 4-Bit	Cohere For AI	CC-BY-NC-4.0	24 GB
Llama 3 8B Instruct	Meta	Llama 3 Community	18 GB

Tab. 1: Overview of models selected for experiments.

3.2 Hardware

All experiments were conducted on a cluster node equipped with an AMD ThreadRipper 5965WX processor (48 cores), 128GB of RAM, and two NVIDIA RTX A5000 GPUs, each with 24GB of RAM.

3.3 Datasets

For our experiments, we use the following two datasets for evaluating the LLM-based chatbots.

Mohler11 Dataset
The Mohler11 [MBM11] dataset is one of the standard dataset for the evaluation of ASAG systems and consists of questions and answers from an introductory computer science course at the University of North Texas. The questions belong to 10 assignments (1-10) and 2 examinations (11-12). For each answer there exist three grades, the instructor grade, a grade given by the teaching assistant and the average of the two. In accordance to [MBM11] only the average grade is taken into account. The PearsonR correlation between the two graders is $\rho = 0.60$. The assignments were graded on an integer scale from 0 to 5, whereas the examinations were graded on a scale from 0 to 10. For ease of comparison the grades from the examinations are divided by 2. Due to its historic nature, the Mohler11 dataset does not include any rubrics based grading.

Table 2 shows information about the assignments and examinations, the number of questions per assignment and the average grade.

Assignment	Questions	Answers	Avg. Grade
1	7	203	3.91
2	7	210	3.92
3	7	217	4.33
4	5	150	4.05
5	4	112	4.21
6	7	182	4.33
7	7	182	4.21
8	6	162	4.19
9	5	135	4.46
10	7	168	4.29
11	10	300	4.09
12	9	252	4.16
Total	81	2,273	
Average	6.8	189.4	4.17

Tab. 2: Statistics for the Mohler11 Dataset

With an average grade of 4.17 points, the dataset is clearly biased towards the correct answer (as many other grading datasets).

WuS24 Dataset

The WuS24 [MH24] dataset consists of questions and answers from an introductory course on probability theory and statistics (Wahrscheinlichkeitstheorie und Statistik), taught in German at Hochschule Bonn-Rhein-Sieg. It includes three questions, each with twenty to fifty graded responses evaluated according to detailed rubrics. These rubrics are structured as bullet points of binary questions, which can be answered with yes or no, with corresponding points (0.5 or 1) assigned to each bullet point. The questions were overall graded on a scale from 0 to 5.

Question	Answers	Avg. Grade
1	20	3.75
2	50	3.79
3	20	3.60
Total	90	
Average	30	3.71

Tab. 3: Statistics for the WuS24 Dataset

The first question from the dataset, accompanied by its corresponding rubrics, is presented in Figure 3.

3.4 Prompting

In our experiments, we make use of the three common types of promts of instruction following LLMs:

System Prompts define the role, behavior and rules for the chat assistant. They can also provide context and guidelines for the interaction. Example: *"You are a helpful assistant that provides accurate information and remains polite."*

User Prompts represent the inputs or questions from the end user. Example: *"What is the role of a prototype program in problem solving?"*

Assistant Prompts contain the responses generated by the LLM based on the system and user prompts. Example: *"The role of a prototype program in problem solving is to simulate the behavior of portions of the desired software product."*

A *Mistral 7B Instruct* does not have a system prompt, we include the system prompt in the first user prompt.

For few-shot prompting we provide the examples in the assistant prompt. The prompt templates used for the Mohler11 and WuS24 datasets can be seen in Figure 1.

Mohler11 dataset

```
{'role': 'system',
 'content':
  'You are helping me grade student answers
   worth 5 points. You answer in the format:
   Grade: [points] / 5'},

// opt few-shot loop:
{'role': 'user',
 'content':
  'Question:\n{question}\n
   Reference Answer:\n{reference_answer}\n
   Student Answer:\n{student_answer}'},
{'role': 'assistant',
 'content': 'Grade: {score} / 5'},

// final prompt
{'role': 'user',
 'content':
  'Question: {question}\n
   Reference Answer: {reference_answer}\n
   Student Answer: {student_answer}'},
```

WuS24 dataset

```
{'role': 'system',
 'content':
  'Du hilfst mir Studierendenantworten zu bewerten.
   Die maximale Punktzahl ist {max_score}.\n\n
   Frage:\n{question}\n\n
   Bewertungsschema:\n{rubrics_with_score}\n\n
   Deine Ausgabe:\n
   1. Fülle das Bewertungsschema aus. (Ja/Nein) für
      jeden Stichpunkt.\n
   2. Berechne die Punktzahl und gebe Sie im Format
      "Punktzahl: [Deine Punktzahl] / 5" aus.\n'},

// opt few-shot loop:
{'role': 'user',
 'content':
  'Frage: {question}\n
   Studierendenantwort:\n{answer}\n
   Bewerte diese Antwort.'},
{'role': 'assistant',
 'content':
  'Bewertungsschema:\n{rubrics}\n\n
   Punktzahl: {score} / 5'},

// final prompt
{'role': 'user',
 'content':
  'Frage: {question}\n
   Studierendenantwort:\n{answer}\n
   Bewerte diese Antwort.'},
```

Fig. 1: Prompts for the Mohler11 and WuS24 dataset. The user and assistant prompts are repeated for one- and two-shot prompting, and left out for zero-shot.

3.5 Evaluation

Reproducibility of LLM Grades
In our first experiment, we investigate how the sampling temperature and the number of examples shown to the model influence the reproducibility of the generated grades.

To do this, we selected two questions from the Mohler11 dataset (1.2 and 5.3), which together include a total of 51 answers. Each model graded each answer 10 times. We repeated this process for four temperatures (0.0, 0.3, 0.5, and 0.7) and tested three scenarios for the number of examples shown to the model (zero-shot, one-shot, and two-shot). For each answer we calculate the standard deviation among the 10 runs. Then we report the distribution of the standard deviation across all answers.

The results in Figure 2 show, that even when setting the temperature to zero, it is evident that *GPT 3.5 Turbo* and *Llama 3 8B Instruct* do not consistently produce identical outputs. We investigated the logits of *Llama 3 8B Instruct* and found that in those cases the model assigns the same score to multiple output tokens.

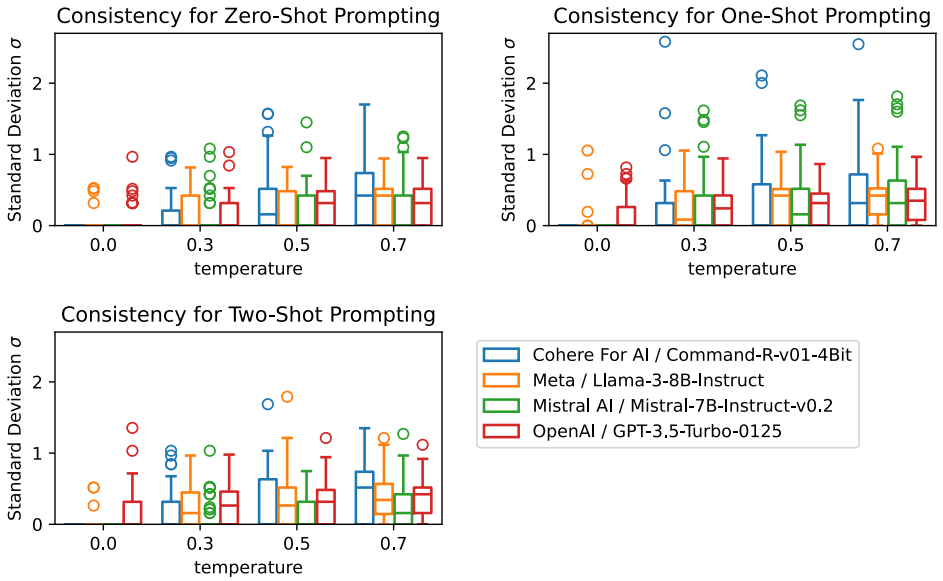


Fig. 2: Reproducibility of grades generated by models across different scenarios on questions 1.2 and 5.3 of the Mohler11 dataset.

Reference Answer Based Grading
In our second experiment, we investigate how the number of examples shown to the

model influences the generated grades. All models evaluated all Mohler11 answers with a temperature of 0.0, testing zero-shot, one-shot, and two-shot scenarios.

The best overall performance (RMSE) is achieved by *GPT 3.5 Turbo* when grading given one example, as can be seen in Table 4.

Model	Zero-Shot	One-Shot	Two-Shot
GPT 3.5 Turbo	**1.21** (0.67)	**1.12** (0.69)	**1.15** (0.71)
Llama 3 8B Instruct	1.44 (0.64)	1.26 (**0.71**)	1.36 (0.70)
Command R v01 4-Bit	1.45 (0.61)	1.35 (0.62)	1.28 (0.65)
Mistral 7B Instruct v0.2	1.34 (**0.69**)	1.26 (0.70)	1.16 (**0.73**)

Tab. 4: RMSE (PearsonR ρ) on Mohler11 Dataset. Using zero-shot prompting, Mistral 7B failed to provide a single grade as a response in 0.47% of cases. In these cases we assigned the maximum error of 5.

Adherence to Grading Rubrics

Finally, we also evaluated the model's ability in grading German responses and adhering to grading rubrics using the WuS24 dataset. In our evaluation, we tested the model with a temperature of 0.0 under various conditions: no examples provided (zero-shot), one example (one-shot), and two examples (two-shot). Evaluation methods encompassed (A) grading solely based on the reference answer, (B) prompting the model to fill out rubrics and generate a grade, and (C) letting the model complete the rubrics, after which we apply our grading scheme by algorithmically parsing the model outputs. An example of a one-shot conversation is presented in Figure 4.

The results are presented in Table 5. It can be seen, that in almost every experiment the models perform best when grading according to rubrics. *GPT 4o* performs best for one-shot and two-shot grading, while *Llama 3 8B Instruct* performs best for zero-shot grading.

3.6 Discussion

The experiment on reproducibility of grades, shown in Figure 2, suggests that some models still generate different results even with a temperature of 0.0. Furthermore, providing the LLM with only one example surprisingly seems to decrease consistency in the responses. In contrast to this, we notice that our RMSE (lower is better) and Pearson correlation (higher is better) improve with one-shot compared to zero-shot learing for all used models.

Finally, it can be seen, that in almost every experiment the models perform best when grading according to rubrics. Inspection of the results shows that the algorithmic parsing of the LLM graded rubrics seems problematic, especially for cases in which the models omit parts of the rubrics.

Model	Zero-Shot			One-Shot			Two-Shot		
	A	B	C	A	B	C	A	B	C
GPT 4o	1.17	0.93	1.09	**0.85**	**0.75**	**0.69**	**0.83**	**0.74**	**0.73**
GPT 3.5 Turbo	1.45	1.86	1.91	0.97	1.08	0.99	0.91	1.12	1.10
Llama 3 Instruct	**0.84**	**0.79**	**0.80**	1.10	1.05	1.04	0.94	0.98	0.95
Command R v01 4-Bit	1.31	1.42	1.94	1.05	0.84	0.91	0.97	0.88	0.99
Mistral 7B Instruct v0.2	1.07	2.09	2.06	1.13	1.04	0.75	1.31	0.95	0.97

Tab. 5: RMSE on the WuS24 dataset. Column A shows the results when grading is based solely on the reference answer. Column B presents the outcomes when the model fills out rubrics and assigns grades accordingly. Column C displays the results when the model completes the rubrics, and the rubric point grading scheme is subsequently algorithmically applied to these rubric entries by parsing the model outputs. Using zero-shot prompting with rubrics, Command R v01 failed to provide a single grade in two instances. We assigned the maximum error of 5 for these cases.

4 Conclusion and Future Work

In this study, we examined the effects of few-shot prompting, the temperature parameter and the impact of rubrics on automated short answer grading using LLMs. Overall, we see that these models perform well without rubrics on the Mohler11 dataset. On the WuS24 dataset, rubric-based grading outperforms reference answer-based grading, offering better grade explainability and serving as feedback for students. However, while we already find these LLM based grading results very promising, they are not perfect. Hence, we suggest using rubrics for semi-automated pre-grading, followed by human conversion of rubrics into grades. In the future, we aim to investigate few-shot approaches in more depth by providing more examples and performing cross-validation across all graded answers.

Acknowledgements

This work was supported by the MKW NRW JupyterHub.nrw project.

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Appendix

Erläutern Sie kurz jeweils den Mittelwert, Median und Modus und gehen dabei insb. auch auf mögliche Unterschiede ein. Welches der Maße würden Sie bei großer Varianz verwenden und warum?

- Wird der Mittelwert klar verständlich erklärt? (Ja/Nein), 1 Punkt
- Wird darauf hingewiesen, dass der Mittelwert anfällig für Ausreißer ist? (Ja/Nein), 0,5 Punkte
- Wird der Median klar verständlich erklärt? (Ja/Nein), 0,5 Punkte
- Wird erklärt, dass der Median bei einer Stichprobe mit gerader Anzahl als Mittelwert der zwei mittleren Werte berechnet wird? (Ja/Nein), 0,5 Punkte
- Wird darauf hingewiesen, dass der Median robust gegenüber Ausreißern ist? (Ja/Nein), 0,5 Punkte
- Wird der Modus klar verständlich erklärt? (Ja/Nein), 1 Punkt
- Wird die Mehrdeutigkeit des Modus bei mehreren häufigsten Werten erläutert (bi--modal/multi-modal)? (Ja/Nein), 0,5 Punkte
- Wird plausibel erklärt, warum bei großer Varianz der Median bevorzugt werden sollte? (Ja/Nein), 0,5 Punkte

Briefly explain the mean, median and mode, focusing especially on possible differences. Which measure would you use for a dataset with high variance and why?

- Is the mean clearly explained? (Yes/No), 1 point
- Is explained that the mean is sensitive to outliers? (Yes/No), 0.5 points
- Is the median clearly explained? (Yes/No), 0.5 points
- Is explained that the median is calculated as the mean of the two middle values in a dataset with an even number of data points? (Yes/No), 0.5 points
- Is explained that the median is robust against outliers? (Yes/No), 0.5 points
- Is the mode clearly explained? (Yes/No), 1 point
- Is the ambiguity of the mode with multiple most frequent values explained (bi-modal/multi-modal)? (Yes/No), 0.5 points
- Is explained why the median should be preferred in the case of high variance? (Yes/No), 0.5 points

Fig. 3: Rubrics for Question 1 of the WuS24 Dataset. Left side (used for the experiments): The original German version. Right side: English translation for the reader

SYSTEM:
Du hilfst mir Studierendenantworten zu bewerten. Die maximale Punktzahl ist 5.
Frage: Mittelwert, Median und Modus
Erläutern Sie kurz jeweils den Mittelwert, Median und Modus und gehen dabei insb. auch auf mögliche Unterschiede ein. Welches der Maße würden Sie bei großer Varianz verwenden und warum?
Bewertungsschema:
- Wird der Mittelwert klar verständlich erklärt? (Ja/Nein), 1 Punkt
- Wird darauf hingewiesen, dass der Mittelwert anfällig für Ausreißer ist? (Ja/Nein), 0,5 Punkte
- Wird der Median klar verständlich erklärt? (Ja/Nein), 0,5 Punkte
- Wird erklärt, dass der Median bei einer Stichprobe mit gerader Anzahl als Mittelwert der zwei mittleren Werte berechnet wird? (Ja/Nein), 0,5 Punkte
- Wird darauf hingewiesen, dass der Median robust gegenüber Ausreißern ist? (Ja/Nein), 0,5 Punkte
- Wird der Modus klar verständlich erklärt? (Ja/Nein), 1 Punkt
- Wird die Mehrdeutigkeit des Modus bei mehreren häufigsten Werten erläutert (bi-modal/multi-modal)? (Ja/Nein), 0,5 Punkte
- Wird plausibel erklärt, warum bei großer Varianz der Median bevorzugt werden sollte? (Ja/Nein), 0,5 Punkte
Deine Ausgabe:
1. Fülle das Bewertungsschema aus. (Ja/Nein) für jeden Stichpunkt.
2. Berechne die Punktzahl und gebe Sie im Format "Punktzahl: [Deine Punktzahl] / 5" aus.

USER:
Frage: Mittelwert, Median und Modus
Erläutern Sie kurz jeweils den Mittelwert, Median und Modus und gehen dabei insb. auch auf mögliche Unterschiede ein. Welches der Maße würden Sie bei großer Varianz verwenden und warum?
Studierendenantwort:
Mittelwert gibt den Durchschnitt aller einbezogenenen werte an. Median gibt den in der Mitte einer sortieren Liste sitzenden Wert an. Modus ist der Wert der in der Liste am häufigsten vorkommt. Den Mittelwert, denn die Varianz gibt die mittlere quadratische abweichung um ihren Mittelwert an.
Bewerte diese Antwort.

ASSISTANT (given as example):
Bewertungsschema:
- Wird der Mittelwert klar verständlich erklärt? (Ja)
- Wird darauf hingewiesen, dass der Mittelwert anfällig für Ausreißer ist? (Nein)
- Wird der Median klar verständlich erklärt? (Ja)
- Wird erklärt, dass der Median bei einer Stichprobe mit gerader Anzahl als Mittelwert der zwei mittleren Werte berechnet wird? (Nein)
- Wird darauf hingewiesen, dass der Median robust gegenüber Ausreißern ist? (Nein)
- Wird der Modus klar verständlich erklärt? (Ja)
- Wird die Mehrdeutigkeit des Modus bei mehreren häufigsten Werten erläutert (bi-modal/multi-modal)? (Nein)
- Wird plausibel erklärt, warum bei großer Varianz der Median bevorzugt werden sollte? (Nein)
Punktzahl: 2.5 / 5

USER:
Frage: Mittelwert, Median und Modus
Erläutern Sie kurz jeweils den Mittelwert, Median und Modus und gehen dabei insb. auch auf mögliche Unterschiede ein. Welches der Maße würden Sie bei großer Varianz verwenden und warum?
Studierendenantwort: Mittelwert: Ist der Durchschnitt aller Werte. Er wird berechnet indem man ale Werte addiert und das Ergebnis dann durch die Menge aller Werte teilt. Median: Ist das Element was genau in der Mitte eines sortierten Datensatzes steht. Modus: Ist das Element was am häufigsten in einem Datensetz vorkommt. Was davon bei großer Varianz? Dabei sollte man den Median nehmen da dieser anders als der Mittelwert nicht bei großen Varianzen stark abweicht.
Bewerte diese Antwort.

ASSISTANT:
Bewertungsschema:
- Wird der Mittelwert klar verständlich erklärt? (Ja)
- Wird darauf hingewiesen, dass der Mittelwert anfällig für Ausreißer ist? (Nein)
- Wird der Median klar verständlich erklärt? (Ja)
- Wird erklärt, dass der Median bei einer Stichprobe mit gerader Anzahl als Mittelwert der zwei mittleren Werte berechnet wird? (Nein)
- Wird darauf hingewiesen, dass der Median robust gegenüber Ausreißern ist? (Nein)
- Wird der Modus klar verständlich erklärt? (Ja)
- Wird die Mehrdeutigkeit des Modus bei mehreren häufigsten Werten erläutert (bi-modal/multi-modal)? (Nein)
- Wird plausibel erklärt, warum bei großer Varianz der Median bevorzugt werden sollte? (Ja)
Punktzahl: 4 / 5

Fig. 4: Example of a conversation using One-Shot Prompting on the WuS24 dataset with GPT-3.5 Turbo and rubrics. The first assistant prompt is provided as an example to the model.